

Swiss Enterovirus Symposium

Abstract Book



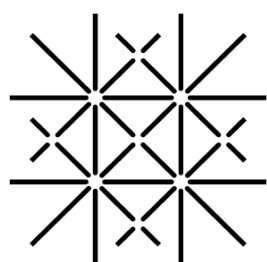
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European Non-Polio Enterovirus Network



**University
of Basel**

Keynote Lectures



Keynote Lectures

Keynote I – Heli Harvala, Enteroviruses – Harmless Fellow Passengers or the Next Disease X?

University of Turku and Turku University Hospital, Finland; co-founder of ENPEN

This opening keynote will provide a brief background to enteroviruses, and their clinical significance. The talk will introduce the use of clinical, environmental and genomic surveillance utilized to obtain a better understanding of their clinical significance, possible disease burden and likely future emergence.

Keynote II – Javier Martin, Poliovirus and Beyond: Enterovirus Genomics in Wastewater

National Institute for Biological Standards and Control (NIBSC), UK

Wastewater surveillance provides a powerful window into enterovirus circulation, revealing transmission that may remain largely invisible to clinical surveillance. Poliovirus offers the clearest demonstration of its public-health value: environmental detection can identify silent circulation, provide early warning of importations and outbreaks, and support the genetic reconstruction of transmission events during the eradication endgame. This keynote will examine how combining wastewater surveillance with clinical sampling and advanced genomic analysis can deepen our understanding of enterovirus epidemiology and evolution. Poliovirus examples will illustrate how whole-genome sequencing can extend conventional VP1 analysis by identifying recombination, resolving relationships between apparently distinct transmission chains and providing additional insight into the origins and spread of vaccine-derived polioviruses. The same integrated approach has wider relevance for non-polio enteroviruses. EV-D68 will be presented as an important case study in which parallel analysis of wastewater and clinical sequences can reveal otherwise undetected circulation, capture greater viral diversity and place viruses associated with respiratory and neurological disease within their broader population context. Together, these findings support a transition from pathogen-specific surveillance towards integrated enterovirus surveillance, linking environmental and clinical data with whole-genome sequencing to detect emerging threats, interpret their evolution and strengthen public-health preparedness.

Keynote III – Katharina Röltgen, Reading infection histories from antibodies: longitudinal serology from SARS-CoV-2 to enteroviruses

Swiss Tropical and Public Health Institute (Swiss TPH), Switzerland

Repeated respiratory virus infections occur throughout life, yet their long-term infection and immune dynamics remain poorly understood. Longitudinal serology can help reconstruct individual infection histories from antibody peaks while revealing antibody boosting, waning, and persistence over time. Long-term biobanks with serial samples are rare, but longitudinal HIV cohorts provide an opportunity through regular clinical follow-up and repeated sample collection. We leverage two exceptional cohorts of people living with HIV: the Swiss HIV Cohort Study (SHCS) and the Kilombero and Ulanga Antiretroviral Cohort (KIULARCO) in Tanzania, both combining long-term follow-up, biannual plasma sampling, and detailed clinical information. We first analyzed serial plasma samples from 35 SHCS participants with preserved CD4 cell counts and HIV viral suppression, followed over 10 years. SARS-CoV-2 antibody patterns, based on differential responses to Spike and Nucleocapsid antigens, demonstrate how infection and vaccination events can be distinguished while revealing antibody boosting and waning. We then expanded this longitudinal approach to IgG responses against influenza viruses, respiratory syncytial virus, parainfluenza viruses, human metapneumovirus, and enteroviruses EV-D68 and EV-A71. Longitudinal enterovirus serology revealed distinct antibody peaks consistent with EV D68 infection, independently confirmed by live-virus neutralization assays. However, EV-D68 peaks sometimes coincided with EV-A71 responses, illustrating how antigenic relatedness and potential cross-reactivity can complicate the attribution of antibody responses to individual viruses, a challenge that extends beyond enteroviruses. To improve specificity, we are now using phylogenetic analyses to identify antigenic regions conserved within, but distinct between, enterovirus species A, B, C, and D as candidates for species-specific serological assays. We will next establish these assays and extend the analyses to KIULARCO, enabling comparison across two distinct epidemiological settings. Combining longitudinal cohorts with species-specific serological tools could provide a framework for reconstructing infection histories and studying humoral immunity over time, with potential application across a broad range of respiratory viruses.

Keynote IV – Kevin Messacar, A Two-Way Lens: Clinical Insights Into and From Multimodal Enterovirus Surveillance

University of Colorado School of Medicine, USA

This talk highlights on how clinical insights guide multimodal surveillance systems, providing further insight into clinical care. With a focus on EV system specifics and data.

Keynote V – Florian Zwagemaker, Ten Years of Modernizing Pathogen Data Infrastructure: How Scalable Platforms Enable Enterovirus Surveillance

National Institute for Public Health and the Environment (RIVM), Netherlands

Genomic surveillance has become central to public health virology, yet its efficacy and impact depend heavily on the underlying data infrastructure. This critical foundation is however commonly overlooked when considering the analytical and operational possibilities of pathogen genomics at scale. In this talk, we present a decade-long trajectory of modernizing pathogen data infrastructure at the Dutch National Institute for Public Health and the Environment (RIVM). We trace the technological and organizational steps that enabled this transformation and demonstrate how scalable platforms can support the molecular surveillance of enteroviruses at a national level. Central to this development lies the design philosophy wherein infrastructure and workflows are built under the assumption that every incoming sample may be sequenced, rather than treating sequencing as an auxiliary process reserved for epidemiological outbreak investigations. This approach requires genomics to be properly integrated throughout routine laboratory, data-processing, and surveillance workflows. The resulting transformation spans the expansion of compute and storage capacity at the datacentre level, the development of Gen-EpiX, an open-source, purpose-built, pathogen-agnostic national genomic epidemiology platform, the replacement of end-of-life systems with modern alternatives such as Loculus, and the introduction of browser-based bioinformatics tooling for complex curation and ad-hoc analysis. Using enterovirus surveillance as a case study, we discuss how sustained investment in modern, robust, and interoperable genomic infrastructure collectively strengthens national enterovirus surveillance. Together, these developments provide the scalability, reproducibility, and transparency required to identify, investigate, and respond to viral outbreaks.

Lecture Sessions



Lecture Session I – Epidemiology & Surveillance I

1. Global epidemiology and genetic evolution of enterovirus D68 over a decade (2015-2024): a systematic review

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Enterovirus D68 (EV-D68) is a persistent global health concern, causing mild to severe respiratory disease and, in rare cases, acute flaccid myelitis (AFM), a polio-like paralytic syndrome. Following the pivotal EV-D68 outbreak in 2014, EV-D68 has continued to circulate and evolve worldwide. This systematic review synthesizes global data on epidemiology and genetic evolution of EV-D68 available from 2015-2024.

PubMed was searched for reports published from January 1st, 2015, to October 21st, 2024. Of 722 articles, 220 were eligible for inclusion. Data was analyzed by year and world region. For phylogenetic analysis, a time-scaled phylogenetic tree of all available viral protein 1 sequences from 1990 to October 28th, 2025, was downloaded from Nextstrain. Between 2015 and

2024, 8184 EV-D68 cases and 267 EV-D68-associated AFP cases were reported by 44 and 26 countries, respectively, across all continents. Biennial upsurges followed a distinct even-year pattern until 2020 after which circulation showed geographical diversification.

EV-D68 remained associated with severe respiratory disease and AFM, primarily in children under the age of five, but increasingly in adults. EV-D68 exhibited a ladder-like evolutionary pattern, With subgenotypes B3 and A2/D predominating since 2016 and 2018, respectively, and rapidly branching into novel variants after 2020. The A2/D subgenotype was more frequently detected in adults. Findings reveal the sustained global circulation of EV-D68, its swift genetic evolution and retained

ability to cause severe respiratory disease and neurological complications in a growing demographic. These stress the need for enhanced EV-D68 surveillance, improved diagnostic strategies, and targeted interventions to mitigate EV-D68 morbidity.

2. Reconstructing enterovirus D68 clade-specific transmission dynamics using Bayesian phylodynamics and association with acute flaccid myelitis cases

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Background: Enterovirus D68 (EV-D68) has emerged as a major cause of respiratory disease since the late 2000s, causing large outbreaks across North America and Europe. These outbreaks have also been temporally associated with acute flaccid myelitis (AFM), a rare but severe neurological condition. However, the transmission dynamics of different EV-D68 lineages and their relationship to AFM remain poorly understood.

Methods: We analysed publicly available VP1 gene sequences from EV-D68 isolates collected in North America and Europe.

Using a birth–death phylodynamic framework, we reconstructed changes in transmission through time for two major evolutionary lineages of the virus, termed the A+D and B+C superclades (made up of the A+D and B+C clades respectively). These reconstructions were used to identify historical outbreak periods and compare viral transmission dynamics with reported AFM incidence. **Results:** The inferred transmission histories recapitulate the well-described pattern of large EV-D68 outbreaks since 2014, including disruption of the biennial epidemic cycle during the COVID-19 pandemic and its subsequent re-emergence. Distinct patterns of circulation were observed between the two superclades, with outbreaks often dominated by a single lineage. Comparison with AFM surveillance data suggests that increases in AFM incidence are more closely aligned with transmission of the B+C superclade than the A+D superclade, supporting previous observations linking clade B viruses to neurological disease.

Conclusions: Phylodynamic analysis of EV-D68 genomic data provides valuable insights into the epidemiological history of this emerging pathogen and enables inference during periods of sparse surveillance. Our results support a stronger association between the B+C superclade and AFM and highlight the importance of continued genomic surveillance to monitor the evolution and public health impact of EV-D68

3. Under the Radar: Wastewater Surveillance Reveals Continuous EV-D68 Circulation in Israel, 2024–2025

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Background: Enterovirus D68 (EV-D68) has emerged globally as an important respiratory pathogen associated with severe respiratory illness and acute flaccid myelitis (AFM). Since the large multinational outbreak in 2014, recurring outbreaks have been reported worldwide. In Israel, a retrospective investigation identified an EV-D68 outbreak during 2021, demonstrating a strong correlation between clinical cases and wastewater viral loads. However, routine clinical testing for EV-D68 is not performed, and its long-term circulation patterns remain poorly understood.

Methods: As part of the Israeli national environmental surveillance program,

wastewater samples were collected from multiple wastewater treatment plants between 2024 and 2025 and analyzed using a specific RT-qPCR assay for EV-D68. Viral dynamics were evaluated over time and compared with respiratory HMO clinical surveillance and hospitalized patients data tested for EV-D68.

Results: EV-D68 RNA was detected in wastewater throughout the study period, indicating continuous circulation within the Israeli population.

Viral concentrations demonstrated a reproducible seasonal pattern, with increasing levels during autumn and lower levels during spring and early summer. Clinical surveillance identified EV-D68-positive respiratory samples throughout the study period, with approximately 1% positivity among tested patients, supporting the wastewater findings and indicating persistent community circulation. These observations suggest that EV-D68 remains endemic in Israel even outside recognized outbreak periods.

Conclusions:

Environmental surveillance revealed sustained EV-D68 circulation in Israel and identified seasonal increases that may not be fully captured by routine clinical diagnostics. Wastewater-based epidemiology provides a sensitive population-level tool for monitoring EV-D68 dynamics and may serve as an early warning system for future outbreaks and the emergence of severe disease-associated strains.

Lecture Session II – Epidemiology & Surveillance II

1. From local outbreak to international insight: detecting and managing emerging enterovirus threats through England's integrated surveillance system

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Background:

Integrated enterovirus (EV) surveillance is essential for identifying changes in epidemiology and supporting outbreak investigation. We describe the application of England's national surveillance system during an outbreak of EV E17.

Methods:

England's EV surveillance system integrates voluntary local testing with molecular reference laboratory data from primary and secondary care. After clinicians at a central London hospital reported suspected EV infections in infants from their catchment, analysis of national surveillance data identified an E17 outbreak between February and April 2026. Cases were linked to secondary care data for diagnostic and severity information.

Results:

We identified 18 laboratory-confirmed E17 cases (16 in London and 2 in Manchester), and 3 additional suspected cases in London. There was no substantial increase in other EVs in these areas.

The median case age was 40 days; most (14/18; 77.8%) were aged ≤ 3 months and 5/18 (27.8%) were neonates. Most were male (13/18; 72.2%) and reported by the same hospital (12/18; 66.7%). E17 was detected most commonly through cerebrospinal fluid (CSF) testing (n=11), through stool testing (n=5) or both (n=2). This noted increase in E17 detections coincided with a similar emergence of E17 observed in New York and Israel. Cases predominantly presented with non-specific febrile illness. Two cases presented with meningitis. Of the 18 confirmed cases, 16 (88.9%) were admitted to hospital, with an average length of stay of 2.78 days (range: 1 to 8).

Conclusions:

This investigation demonstrated the value of integrated surveillance for identifying and characterising rare outbreaks. Combining laboratory, clinical and local data enabled timely assessment of E17 transmission and severity and informed public health response. The rarity of the E17 strain, together with information about the community affected, prompted international collaboration to further characterise the strain and investigate its origin.

2. Culture-based wastewater surveillance reveals successive epidemic waves of infectious enterovirus genotypes in Milan, Italy, 2023–2026

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Background

Wastewater surveillance is a valuable tool for monitoring enterovirus (EV) within communities. Coupled with virus isolation in cell culture, followed by molecular characterization, it can detect and type infectious, replication-competent EV, rather than just viral genomic material. This study investigated the diversity and temporal patterns of EV circulating in Milan through wastewater surveillance.

Methods

Wastewater samples were collected weekly from in Milan (population approximately 1.25 million) between January 2023 and May 2026. The samples were concentrated using (PEG 8000)-based method and inoculated onto cell lines (RD, L20B). RNA was extracted from surnatants of cultures showing cytopathic effects and a fragment of the VP1 gene (nt 2602–2950) was sequenced using Sanger method.

Results

A total of 377 wastewater samples were analyzed, 211 (56.0%) yielded EV isolates in RD cell cultures. Sequencing was successfully performed on 65% of the isolates, identifying 16 types belonging to species B and D. Echovirus 6 (E6) was the most frequently identified type (30.7%), followed by E3 (24.1%) and E11 (22.6%). Genotype-specific

epidemic waves were observed: E11 prevailed from mid-April to end of July 2023 and from October 2025 to Mid-March 2026; E6 predominated from Mid-September to the end of November 2023 and from Mid-April to Mid-August 2024, E3 from late 2024 to mid-July 2025.

Conclusions

Combining wastewater surveillance, cell culture isolation, and sequence-based typing enabled monitoring of infectious EV circulating in the community. By targeting viruses recovered in cell culture, this approach identifies active infectious strains and avoids some limitations of direct nucleic acid detection.

Wastewater surveillance revealed diverse circulating EV and successive genotype-specific epidemic waves. These results demonstrate its value for tracking transmission dynamics and enabling early detection of public health-relevant EV.

3. Genetic origins and circulation of emerging EV-A71 recombinant lineages in Europe, 2014–2024

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Background Genetic recombination is a major driver of RNA virus evolution. In enteroviruses (EVs), recombination

happens most often in the non-structural genome, around the 5'UTR, or within the P2 and P3 regions, while the capsid/P1 region is usually much more conserved. Recently, we identified 9 recombinant forms (RFs) in patients infected with EV-A71 C1, reported in 2016–2022. The aim of the present study was to review the existing EV-A71 and other EV-A genomes to identify all RFs and investigate which recombination partners were involved in the emergence EV-A71 RFs. **Methods** We selected 637 clinical specimens from patients with laboratory confirmed EV-A infections reported between 2014 – 2022 (France) and subjected them to whole EV genome sequencing. GenBank was gathered to collect all publicly available EV-A genomes. All EV genomes were analysed to detect recombination breakpoints and identify RFs in each EV type. **Findings** We determined 348 complete EV-A genomes in 8 EV-A types cocirculating

with EV-A71 in France: CVA2 (n=21 sequences), CVA4 (33), CVA5 (26), CVA6 (164), CVA8 (6), CVA10 (47), CVA12 (1), CVA16 (50). We curated a dataset of 4,335 genomes retrieved from GenBank, comprising 1892 EV-A71 sequences, which included 153 sequences from our earlier study. We identified a total of 48 RFs in EV-A71 and classified them as circulating or sporadic (unique). EV-A71 C1 RFs were mosaics arising from multiple recombination events affecting all genomic regions except P1 and involving different EV-A types. **Conclusions** Recombination helps EV-A71 generate genetic diversity by moving non-structural gene modules between EV-A lineages. Recombination events likely played important roles in the emergence of EV-A71 (sub)genotypes. The multi-breakpoint genomes support the idea that recombination repeatedly reshape the population structure of circulating and epidemic EV-A71.

Lecture Session III – Bioinformatics & Tools

1. EVOLVE: Enterovirus OnLine tools for Viral Epidemiology

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Introduction: Enteroviruses (EV) are a genetically diverse group of RNA viruses associated with a wide spectrum of clinical manifestations. Genomic surveillance is essential to monitor viral evolution, emergence, and transmission dynamics. However, fragmented data sharing across institutions and platforms continues to limit large-scale analyses and hampers coordinated public health responses.

Methods: We developed EVOLVE (<https://evolve.vhir.org>), an open-access, collaborative platform for EV genomic surveillance and sequence characterization. EVOLVE accepts both raw sequencing data (FASTQ, currently Illumina) and consensus sequences (FASTA) and integrates automated Nextflow-based bioinformatics pipelines covering quality control, genome assembly, genotyping, and mutation annotation, enabling end-to-end NGS

analysis within a single environment. Structured metadata fields are embedded throughout, linking genomic data to clinical and epidemiological context.

Results: EVOLVE centralises EV genomic sequences and associated metadata from clinical laboratories, surveillance networks, and research groups into a scalable, interoperable repository. The combination of data storage, sharing functionalities, and embedded analytical pipelines streamlines sequencing workflows and reduces the bioinformatics barrier for users without bioinformatic background. Early implementation has demonstrated improved accessibility to sequence characterization tools and facilitated downstream analyses.

Conclusions: EVOLVE offers an integrated solution for EV genomic surveillance, bridging the gap between sequencing data generation and epidemiological interpretation, while lowering technical barriers and promoting open data sharing. Broad participation from the scientific and public health communities is essential to maximise its impact and drive continuous improvement of the platform.

2. Enterovirus Recombination: A Comparative Analysis

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Enteroviruses are genetically and antigenically diverse with each species containing several genetically-defined and antigenically-distinct types within it. Previously, it has been suggested that, due to pervasive recombination, enteroviruses may be best conceptualized as collections of capsids and nonstructural proteins that evolve independently. We employed complementary entropy, network, and phylogenetic approaches to quantify the genetic diversity and recombination frequencies of each species. We show that recombination differs substantially between species, with Enterovirus B recombining most frequently followed by Enterovirus A, C, and D. We identify a strong correlation between recombination and the magnitude of difference in diversity between the capsid and nonstructural proteins of a given enterovirus species. Given this finding, we propose a model in which recombination does not simply decouple the evolution of genomic regions, but rather has a profound impact on the diversification patterns of human enteroviruses due to functional constraints on the set of compatible combinations. Our results indicate that antivirals and vaccines targeting nonstructural proteins may be more broadly effective than those targeting the capsid. Furthermore, the existence of higher-order constraints on genome evolution could have wider implications for the diversification of recombining viruses. Currently, we are conducting in vitro experiments to determine the molecular recombination rate of a representative polymerase from each species and elucidate determinants of recombination partner compatibility.

3. Beyond consensus genomes: Insights from poliovirus wastewater surveillance into strain-level resolution of mixed populations

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High-resolution reconstruction of enterovirus outbreak emergence and transmission dynamics relies on accurate discrimination of closely related viral lineages in environmental samples. By capturing viral shedding from both symptomatic and asymptomatic individuals,

environmental surveillance can reveal transmission patterns and genetic diversity that may be missed through clinical sampling alone. However, environmental samples comprise complex mixed viral populations originating from multiple individuals and may contain several closely related lineages. As a result, standard consensus-based bioinformatic

pipelines often fail to resolve lineage-defining mutations, limiting accurate outbreak tracking and phylogenetic reconstruction.

In Israel, poliovirus environmental surveillance is challenged by the simultaneous presence of vaccine strains and emerging vaccine-derived polioviruses (VDPVs). We applied Oxford Nanopore MinION sequencing coupled with novel bioinformatic approaches to

resolve mixtures of closely related poliovirus type 1 strains in environmental samples. This strategy enabled identification of the earliest outbreak-associated strain from a mixed sample dominated by vaccine virus. We further separated outbreak strains from additional mixed samples and demonstrated that the lineage had accumulated sufficient mutations to meet WHO criteria for classification as vaccine-derived poliovirus type 1

(VDPV1).

Our findings demonstrate the value of combining long-read sequencing with tailored bioinformatic analyses to resolve mixed viral populations in environmental samples. This approach enables earlier detection of emerging VDPVs, improves phylogenetic resolution, and supports more informed public health responses during enterovirus outbreaks.

Lecture Session IV – Viral Fitness & Dynamics

1. Deep Sequencing Maps Quasispecies Dynamics and the Fitness Landscape of Poliovirus

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A quantitative understanding of viral evolution requires measuring how mutations accumulate, persist, and are eliminated from populations. However, mutation-accumulation studies have been limited by the resolution of conventional sequencing, which primarily detects high-frequency variants. Here, we examined poliovirus evolution over multiple passages using a high-accuracy sequencing strategy capable of detecting nearly all possible single-nucleotide mutations across the genome. Analysis of mutational frequency spectra over time captured the diversification of neutral and deleterious mutations and revealed convergence toward mutation–selection balance. These dynamics indicate that nucleotide identity is a major determinant of a mutation’s evolutionary fate. Fitness estimates inferred from temporal population dynamics strongly correlated with evolutionary variability across Enterovirus C sequences. This correspondence is comparable to that reported for deep mutational scanning studies, suggesting that high-resolution experimental evolution can recover DMS-like information while also capturing the temporal dynamics of mutation accumulation and purifying selection. Haplotype reconstruction further identified candidate epistatic

interactions, revealing combinations of mutations whose behavior cannot be explained by individual effects alone. Finally, mapping inferred selection onto available poliovirus protein structures showed that constrained residues are enriched in known or predicted functional and structural regions, including sites potentially involved in folding, enzymatic activity, RNA binding, capsid assembly, polyprotein processing, and replication. These assignments should be viewed as hypothesis-generating rather than definitive.

Together, these results provide a quantitative view of how mutation and selection shape viral quasispecies evolution and constrain poliovirus evolutionary trajectories.

2. Targeting rhinovirus IRES as new approach for antivirals

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The Internal Ribosomal Entry Site (IRES) sequence is a conserved RNA element present in the Picornaviridae family that plays a crucial role in the viral life cycle. Targeting the IRES is feasible, as demonstrated by compounds showing activity against EV71 in vitro and in vivo. However, available compounds mostly target stem-loop II, whose relatively simple structure may lead to off-target effects. Furthermore, current IRES

structural models are largely based on computational predictions and have not been extensively validated in infected cells.

To address these limitations, we probed viral RNA directly in infected cells and determined the secondary structure of the rhinovirus 8 (RV8) IRES using DMS-MaP. The experimentally derived structure guided the design of a library of LNA-ASOs (locked nucleic acid antisense oligonucleotides) that either unfold or induce degradation of the target RNA. Among the tested ASOs, four exhibit nanomolar activity against RV8 without detectable toxicity in vitro and in human derived respiratory tissue models. Ongoing studies focus on further optimization of these candidates, including evaluation of viral resistance, synergistic combinations, and activity against additional enteroviruses, including EV68 and EV71.

3. Tracking Enteroviruses Outbreaks Through Environmental Surveillance and Molecular Epidemiology: Polio as Model

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Background

Unlike poliovirus, active non-polio enterovirus (NPEV) surveillance is voluntary and region-dependent, causing incomplete outbreak data and missed transmissions from clinical under-reporting. Standard clinical data fails to capture asymptomatic circulation, vital for understanding true outbreak dynamics and the communities involved. Integrating environmental surveillance (ES) with advanced molecular epidemiology addresses these gaps. We use Israel's 2022–2023 cVDPV2 outbreak as a model demonstrating how this combined approach resolves hidden, asymptomatic enterovirus dynamics.

Methods

Israel's ES network was expanded from 15 routine sites to 45 during the outbreak (April 2022–June 2023), covering ~70% of the population. cVDPV2-positive samples underwent whole-genome sequencing (WGS) and phylogenetic analysis to resolve transmission pathways.

Findings

The ES system detected 256 genetically linked environmental samples. Virus characterization via WGS identified a unique recombination event with an NPEV C species, providing a critical molecular signature that enabled early detection of a cross-border outbreak linking Israel, London, New York, and Canada. Parallel to this global identification, expanded ES enabled targeted tracking of under-monitored areas, revealing that the outbreak

circulated silently, predominantly in Jewish Ultra-Orthodox populations. Finally, phylogenetic analysis resolved three distinct lineages, tracking outbreak dynamics and uncovering epidemiological links and silent transmission pathways between communities.

Conclusion

Integrating expanded ES and WGS offers a powerful, scalable model for

enterovirus monitoring. By mapping asymptomatic circulation and clinical under-reporting through broad genetic signatures and unique recombination events, this framework resolves local transmission pathways and tracks cross-border viral spread, and enhances our ability to identify future outbreaks and predict future transmission routes.

Poster Session



Poster Session

1. Recombination-driven evolution of Coxsackievirus A4 and its links to emerging EV-A71 lineages

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Coxsackievirus A4 (CVA4), a member of the Enterovirus A species (Picornaviridae), is associated with febrile illness in children and is a major cause of herpangina in the Asia-Pacific region. This study aimed to characterise the genetic diversity of CVA4 strains circulating in France and to investigate potential recombination events. Forty-eight clinical samples from patients with laboratory-confirmed CVA4 infection reported in France between 2013 and 2022 were selected for sequencing after PCR-based viral genome enrichment. Newly generated sequences were combined with 180 complete CVA4 genomes available from GenBank. Phylogenetic relationships were inferred using the P1 region, while full-genome analyses were performed to identify recombination breakpoints and define recombinant forms (RFs). Between 2013 and 2022, 212 laboratory-confirmed CVA4 infections were reported in France. Among the 48 clinical samples analysed, 35 complete genomes were obtained, corresponding to a sequencing success rate of 72.9%. P1-based phylogenetic analysis showed that sequences of French isolates clustered within

subclades C1, C3, D2, and E. Genogroup E, composed exclusively of sequences generated from this study, represents a novel lineage. Among the 215 complete CVA4 genomes analysed, 45 RFs were identified, including 22 from French sequences. Notably, two RFs showed phylogenetic relatedness in the P2/P3 regions to recently described EV-A71 RFs. CVA4 circulating in France exhibits substantial genetic diversity, with co-circulating subclades and the emergence of a novel genogroup. The high number of RFs underscores the central role of recombination in CVA4 evolution. The observed relatedness between some CVA4 and EV-A71 RFs suggests that CVA4 may contribute to the emergence of novel EV-A71 lineages in Europe.

2. Whole genome sequencing of the re-emergent Enterovirus Echovirus 17 cluster in New York State in 2026.

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Enteroviruses (EVs), members of the Picornaviridae family, can infect multiple organ systems. Approximately 30-50 million individuals are infected yearly, predominately patients < 1 yr old, with transmission peaking commonly from July – Sep in the US. Echovirus 17 (Ec17) has caused infections in neonates, with reports showing some severe and fatal outcomes. We report the recent

detection of 25 Ec17 cases, the first detection of this genotype in the US in over 30 years. Samples testing positive at hospitals for EVs were submitted to the Wadsworth Center for typing. Confirmation by EV-specific real-time RT-PCR was followed by genotyping with Sanger sequencing of Viral Capsid Protein 1 (VP1) region. Whole genome sequencing (WGS) was done following PCR enrichment with a SISPA protocol. Reads were mapped to the reference sequence (AY302543) and the consensus analyzed. A multiple alignment was performed with CHHE-29, AY302543, and the three sequences obtained from this study. VP1 typing revealed 25 positive Ec17 cases, with a mean age of 28 d, and chief clinical complaint of non-specific febrile illness. VP1 phylogenetic analysis demonstrated five strains with 2-5 bp differences. Three isolates, representing distinct strains, were cultured and confirmed by IFA. WGS showed that samples had between 1427-1464 bp differences compared to the reference strain, and 20-32 differences between each other. Most of the differences from the reference strain were synonymous, with around 6% (~90) causing an amino acid change. This study outlines the first detection of Ec17 in the US in more than 30 years and in a largely neonatal cohort. Surveillance of EV-positive cases enabled detection of this re-emergent genotype, which would have been inapparent through routine clinical diagnostic testing. EV genotyping plays an important role in monitoring outbreaks in the US.

3. Comparative seroprevalence and immune profiling of enteroviruses in European and African populations

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Enteroviruses are common human pathogens and comprise more than 100 serotypes classified into four species (A-D). Among these, EV-D68 and EV-A71 are the best-characterized viruses and have caused major outbreaks. However, population-level seroprevalence of enteroviruses, the extent of cross-reactive protection within and across enterovirus species, infection frequencies, and the durability of long-term immunity are largely unknown. Serological studies provide critical information on population-level virus exposure but are often limited to single time points. In contrast, longitudinal serological studies allow characterization of individual immune profiles that evolved from repeated viral exposures and vaccinations over time, producing long-term seroprevalence data that are crucial for global outbreak preparedness and guiding public health interventions. However, the long-term dynamics of antibody responses to enteroviruses have not been systematically investigated. To address this knowledge gap, we are characterizing humoral immune responses to enteroviruses in sequential samples from individuals in Europe and Africa followed for more than 10 years using antibody-binding and virus neutralization assays. Antibody responses to EV-D68 and EV-A71 are being measured to assess changes in population-level seroprevalence over

the past 15 years. To extend these analyses to all four enterovirus species, we are identifying and recombinantly expressing species-specific antigenic domains based on multiple sequence alignments of publicly available enterovirus sequences. Candidate domains are selected for high within-species conservation and low cross-species similarity to minimize cross-reactive antibody detection. This approach enables species-specific tracking of enterovirus exposure over time and the assessment of immune durability and (re-)infection frequencies at the individual level.

4. Genomic Insights into Community Transmission of Coxsackievirus A24 Variant during the 2024 Acute Hemorrhagic Conjunctivitis Outbreak in Coastal Kenya

John Mwita Morobe^{1,2}, Edidah M. Ong'era¹, Arnold W. Lambisia^{1,2}, Samuel O. Odoyo¹, Martin Mutunga¹, Esther N. Katama¹, Matt Keeling³, Katherine Gallagher^{1,4}, Edward C. Holmes⁵ & **Charles N. Agoti^{1,6}**

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⁶School of Public Health, Pwani University, Kilifi, Kenya.

Coxsackievirus A24 variant (CVA24v), an Enterovirus C, causes large acute haemorrhagic conjunctivitis outbreaks globally, yet its community transmission dynamics remain poorly understood. We performed whole-genome sequencing of longitudinal CVA24v-positive samples to characterize household and community transmission. We leveraged on a cohort study of 714 individuals (67 homesteads; 113 households) within Kilifi County screened for CVA24v during August 2023 - December 2024. Whole-genome sequencing was performed for 217 CVA24v-positive samples (36.3% of 597 confirmed infections) identified. The outbreak was driven by CVA24v genotype IV. The overall study cohort infection rate was 30.5%, with children < 10 years most affected. While the study genomes formed a single monophyletic group, phylogenetic analysis revealed two transmission patterns: 15 homesteads had single viral introductions with rapid within-homestead spread, while 16 experienced repeated invasion by distinct strains. Ancestral state reconstruction estimated 64

transmission events with nearly half identified as community-to-homestead importations, 40.6% inter-homestead spread, and 10.9% exports.

Epidemiological modelling estimated a mean generation time of 6.0-6.9 days. Longitudinal sequencing of 48 individuals, coupled with shedding durations lasting up to 332 days revealed time-dependent viral evolution: minimal change within 7 days (0-1 SNPs) but progressive mutation accumulation beyond 21 days (>5 SNPs). CVA24v homestead/household epidemiology is highly heterogeneous: some homestead had limited within-homestead

transmission following single introductions, while others served as amplification and export hubs. Targeted interventions in high-risk homestead particularly those with young children could interrupt both within- and between-homestead transmission chains.

5. Exploring whole genome sequence methodologies to optimise enterovirus surveillance: a work in progress

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²Antiviral Unit. Virus Reference Department. UK Health Security Agency. London, England.

Enterovirus (EV) surveillance at the Enteric Virus Unit (EVU), UKHSA, is currently based on a real-time PCR confirmatory assay targeting the 5' untranslated region (5' UTR) alongside genotypic characterisation using Sanger sequencing targeting the VP1 gene. A PCR-amplicon approach targeting the P1 genomic region is used for samples that fail initial genotyping however this methodology only sequences half of the EV genome.

The aim of this work is to evaluate a targeted-sequence capture approach to recover the EV full genome. Generation of whole genome sequence data will confirm initial sanger sequencing genotyping results and will allow the analysis of additional sequence data information for phylogenetic and recombination studies.

Using a custom designed EV capture panel and an in-house bioinformatics pipeline, high-quality sequencing data

was generated and analysed. Initial testing included 89 samples from diverse sources (blood, gastrointestinal, neurological, respiratory, swabs and cardiac) from years 2017, 2021, 2022, 2023 and 2024. Genotype assignment for whole genome sequence data generated was confirmed in 82% (n=73) of samples. Quality of sequence data produced on the Illumina MiSeq platform showed consistent read coverage across the entire EV genome (7.5Kb) with a maximum of 2×10^6 reads.

Targeted sequence capture methodology can be used to generate EV whole genome sequence data. This methodology proved to be informative and robust to sequence clinical samples from various sources and after long term storage however the entire process is labour intensive and expensive. Alternative methodologies like targeted PCR amplicon approach are being explored however, the development of a primer scheme capable of amplifying the EV full genome across all described strains has been challenging due to the EV genetic diversity.

6. Enterovirus surveillance in Belgium in 2026: Increased detection and predominance of Echovirus 9

Mandy Bloemen¹, Annabel Rector¹, Marc Van Ranst^{1,2}, Lize Cuypers² and **Elke Wollants**¹

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Human enteroviruses (EVs) cause a wide range of clinical manifestations, ranging from mild respiratory illness to severe infections such as meningitis, myocarditis and acute flaccid paralysis (AFP). Their high genomic diversity and capacity for genetic variation make molecular surveillance essential to monitor circulating enterovirus types and detect emerging epidemiological patterns. Enterovirus-positive clinical specimens, including cerebrospinal fluid, faeces and respiratory samples, were submitted from across Belgium to the National Reference Centre for Enteroviruses for molecular genotyping. Typing was performed using targeted RT-PCR generating ~4 kb amplicons covering the NCR-VP4-VP2-VP3-VP1 region, followed by rapid barcoding and Oxford Nanopore Flongle sequencing on a MinION Mk1C platform. Between January and August 2026, 184 samples were received, of which 162 (88.0%) were successfully assigned to a genotype. During the same period in 2025, 119/136 (87.5%) samples were successfully genotyped, corresponding to a 36.1% increase in genotyped samples in 2026. Fifteen different genotypes were identified. Echovirus 9 was most frequently detected (n=52; 37.7%), followed by coxsackievirus B5 (n=31; 22.5%) and echovirus 6 (n=15; 10.9%). In addition, eight enterovirus D68 and two enterovirus A71 cases were detected. Three AFP cases were investigated, with no poliovirus detected. Two stool samples tested negative for enteroviruses, while one CSF sample was positive for Enterovirus B. Molecular surveillance revealed a substantial increase in genotyped enterovirus samples in Belgium during the first eight months of 2026 compared with 2025. Enterovirus B predominated, with echovirus 9 as the most frequently

detected genotype, highlighting changes in enterovirus circulation and the importance of continued molecular surveillance.

7. Pathoplexus: Building a new kind of pathogen database

Emma Hodcroft^{1,2,3}, Chaoran Chen^{4,3}, Cornelius Roemer^{2,3}, **Anna Parker**^{1,3}, Anderson Brito⁵, George Githinji⁶, Senjuti Saha⁷, Richard Neher^{2,3}, Tanja Stadler^{4,3}, Theo Sanderson⁸, the Pathoplexus Association⁹

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⁷Child Health Research Foundation, Dhaka, Bangladesh

⁸London School of Hygiene and Tropical Medicine, London, UK

⁹Pathoplexus, Basel, Switzerland

Sharing viral genomic data is essential for advancing scientific research and informing public health responses. Although platforms such as the International Nucleotide Sequence Database Collaboration (INSDC) and GISAID facilitate data sharing, they do not fully address several needs of the pathogen genomics community. Persistent concerns about data misuse, “scooping,” and restrictions on reuse from protected repositories highlight the need for more flexible and equitable data-sharing systems. Pathoplexus is a community-driven viral genomics database designed to balance data

accessibility with submitter autonomy. Built using open-source technologies and guided by transparent governance, the platform allows data submitters to temporarily control how their data are used while still enabling rapid access for researchers and public health officials. All Pathoplexus submissions are automatically uploaded to the INSDC—either immediately if fully open, or after one year if initially shared under a restricted-use option. Pathoplexus leverages Loculus, an open-source platform for managing viral sequence databases. Its web interface and API support both interactive exploration and automated analyses. The project operates as a non-profit association governed by an international Executive Board and includes members from 14 countries across five continents, reflecting a commitment to equity and diverse public health priorities. Despite launching only recently, Pathoplexus has already made a measurable impact. As of August 2026, the database contains more than 18,500 directly submitted sequences. Pathoplexus hosts the first available Sudan ebolavirus sequence from the February 2025 outbreak in Uganda and sequences from the 2026 Andes virus and ongoing Ebola Bundibugyo virus outbreaks. In 2025, Pathoplexus also received more mpox sequences than any other database.

8. Phylogenetic Analysis of Non-Polio Enteroviruses to Develop a Scalable Nomenclature System

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Naming matters, especially when viruses evolve faster than our labels. Non-polio enteroviruses (NP-EVs) cause diseases ranging from mild infections to severe neurological illnesses in children. However, their clade nomenclature often reflects historical sampling rather than current evolutionary relationships, which complicates communication and surveillance.

We developed a reproducible, algorithmic framework for NP-EV clade designation based on parameterized phylogenetic criteria. This framework replaces manual naming systems, is transferable across enterovirus species, and remains adaptable to virus-specific dynamics.

To enable its practical use, we implemented the framework in Nextclade and created the first NP-EV datasets for Enterovirus D68, EV-A71, Coxsackievirus A16, and A10. The new nomenclature is provided alongside existing systems to facilitate comparison and community adoption. We also evaluated inferred ancestral sequences as functional references to improve consistency and interpretability. Our work is important for enabling scalable evolutionary analysis and actionable, real-time NP-EV surveillance.

General Information



Programme Overview

1st Swiss Enterovirus Symposium Programme

5-8 October 2026

Day 1 Monday 5 October

07:30	Breakfast
09:00	Introduction
09:10	Keynote II · Javier Martin <i>Poliovirus and Beyond: Enterovirus Genomics in Wastewater</i>
09:45	Lecture Session I · Epidemiology & Surveillance I 1 Aurora Hirvonen 2 Margarita Pons Salort 3 Itay Bar-Or
10:45	Coffee Break
11:15	Keynote III · Katharina Röltgen <i>Reading infection histories from antibodies: longitudinal serology from SARS-CoV-2 to enteroviruses</i>
11:50	Lecture Session II · Epidemiology & Surveillance II 1 Christopher Bell 2 Laura Pellegrinelli 3 Bétéle Siméon Bakyono
12:50	Lunch
13:50	Social Activity
15:30	Registration
18:00	Opening and Welcome
18:15	Keynote I · Heli Harvala <i>Enteroviruses – Harmless Fellow Passengers or the Next Disease X?</i>
19:00	Dinner

Day 2 Tuesday 6 October

07:30	Breakfast
09:00	Introduction
09:10	Keynote II · Javier Martin <i>Poliovirus and Beyond: Enterovirus Genomics in Wastewater</i>
09:45	Lecture Session I · Epidemiology & Surveillance I 1 Aurora Hirvonen 2 Margarita Pons Salort 3 Itay Bar-Or
10:45	Coffee Break
11:15	Keynote III · Katharina Röltgen <i>Reading infection histories from antibodies: longitudinal serology from SARS-CoV-2 to enteroviruses</i>
11:50	Lecture Session II · Epidemiology & Surveillance II 1 Christopher Bell 2 Laura Pellegrinelli 3 Bétéle Siméon Bakyono
12:50	Lunch
13:50	Social Activity
16:00	
16:15	Keynote IV · Kevin Messacar <i>A Two-Way Lens: Clinical Insights Into and From Multimodal Enterovirus Surveillance</i>
16:50	Poster Flash Talks (3 min each) 1 Bétéle Siméon Bakyono 5 Karen Fuentes 2 Arthur H. Totten 6 Elke Wollants 3 Salomé Lochmann 7 Anya Parker 4 Charles N. Agoti 8 Nadia Neuner-Jehle
17:30	Coffee & Poster Session
19:00	Dinner

Day 3 Wednesday 7 October

07:30	Breakfast
09:00	Introduction
09:10	Keynote V · Florian Zwagemaker <i>Ten Years of Modernizing Pathogen Data Infrastructure: How Scalable Platforms Enable Enterovirus Surveillance</i>
09:45	Lecture Session III · Bioinformatics & Tools 1 Alejandra González Sánchez 2 Lauren McCormick 3 Neta Zuckerman
10:45	Coffee Break
11:15	Lecture Session IV · Viral Fitness & Dynamics 1 Denny Trimcev 2 Stefano Giunta 3 Merav Weil
12:20	Lunch
13:20	Social Activity
16:15	
16:30	Coffee & Poster Session
17:30	Breakout Discussions
18:45	Meeting Summary & Closing Remarks
19:15	Closing Dinner

Day 4 Thursday 8 October

07:30	Breakfast
09:00	Introduction
09:15	Workshop (Part 1)
10:45	Coffee Break
11:15	Workshop (Part 2)
13:00	Lunch – close of symposium
14:00	

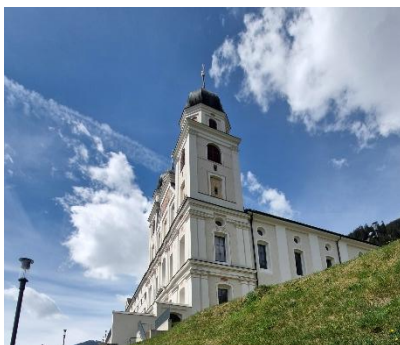
Social Activities

Each afternoon you can choose one of two social activities. Both options run at the same time, so please pick whichever appeals to you most. **You will be asked to select your preferred activity for each day at the conference registration on Monday.** On the day of the activity, simply turn up at the meeting point on time.

Tuesday activities



Village tour: A 90-minute guided walking tour to explore the village of Disentis and learn more about the local mountain culture. To join this tour, please be ready to leave at the monastery entrance at 14:00 sharp!



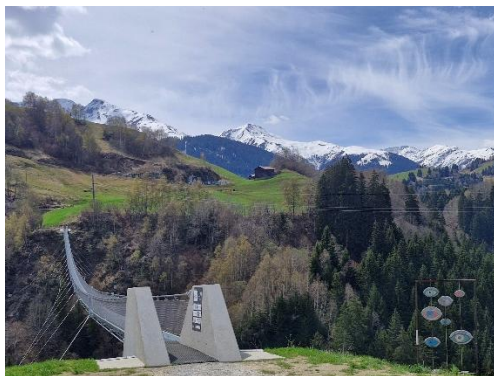
Monastery tour: A guided tour of the Disentis monastery, where we'll learn more about the building and life in the monastery. To join this tour, please be at the tour starting point in the Pfortenplatz (gate square) 14:00 sharp!

Wednesday activities



Caischavedra: The trip to go up Caischavedra mountain by gondola leaves from the Monastery entrance at 13:30 to take the bus (line 420) at 13:40 from the Disentis Claustra bus stop. You are free to stay as long or short as you like, but we recommend taking the 16:00 gondola down at the latest and catch the bus line 491 leaving from 'Acla da Fontauna, center sport' at 16:15. This is the latest connection that will get you back to the monastery in time for the rest of the programme.

IMPORTANT: please ensure to have your travel pass with you.



La Pendentia: Stunning views meets stunning engineering! Take a short, easy hike to the longest suspension bridge in Graubünden: La Pendentia spans 270m across the Vorderrhein river and connects Disentis to the quaint hamlet of Mumpé Medel. To join this tour, please be ready to leave at the monastery entrance at 13:30 sharp! The tour will include:

- Walk to the bridge (25mins there)
- Spend around 15 mins at the bridge
- And then either: walk down to the river, or go and walk around Mumpé Medel (either would be around 30 mins)
- Walk back (25 mins).

Closing Dinner Entertainment



Chiara Jacomet is a versatile artist – a jazz singer and songwriter. Her powerful, dark and velvety voice, together with her expressive stage presence, captivates audiences. In her songs, she gives words to the things that take courage to speak about – to what hurts, what moves us and what is often left unspoken. Her lyrics question social norms and confront the audience with critical questions, inviting listeners to reflect on themselves and the world around them. Singing exclusively in Romansh, she creates a deeply personal and distinctive musical world.

Together with bassist and musician Marius Sommer, Chiara presents her original Romansh compositions in an intimate duo setting. The reduced instrumentation creates space for the interplay between voice and bass, allowing the music to breathe and leaving room for spontaneity, intimacy and intensity. Rooted in jazz, folk influences and lyrical songwriting, their music is both intimate and powerful. The result is a deeply personal and authentic musical experience – tender, honest, poetic and immediate.



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